

Title

Combined Scaling and Photobiomodulation Therapy Augments Arterial Hemodynamic Adaptation in Treated Periodontitis

Abstract

Forty patients with clinically significant periodontal disease underwent randomized allocation to either conventional subgingival instrumentation (mechanical group, $n = 20$) or identical instrumentation supplemented by laser photobiomodulation (combination group, $n = 20$). Periodontal clinical indices improved in both arms over the 5-month observation period; the combination group demonstrated numerically greater gains in gingival health scores (64% improvement versus 41%, $p = 0.06$). Carotid wall hemodynamic assessment revealed divergent trajectories: the combination therapy group exhibited an initial transient 3% decline in shear stress immediately following treatment, followed by sustained 9% elevation at the 20-week timepoint. The mechanical-only group showed minimal net change in arterial hemodynamics across the observation window. Circumferential wall tension and elastic modulus did not differ significantly between groups. These results suggest that photobiomodulation-augmented periodontal care may promote favorable arterial remodeling, potentially contributing to cardiovascular benefit.